Pharmacovigilance Case Assessment

Drug Safety Associate workup | ICH E2A / E2B / E2D, CIOMS, WHO-UMC, Naranjo

Case ID	PV-IN-2026-00417	Prepared	19 Oct 2026
Report type	Spontaneous, healthcare professional, India	Product status	Marketed
Received by MAH (Day 0)	12 Oct 2026	Version	Initial

1. ICSR Validity Check

Minimum criterion (ICH E2D)	Information available	Met?
Identifiable patient	58-year-old male, initials K.S., 82 kg	Yes
Identifiable reporter	Treating physician, internal medicine, Pune, India	Yes
Suspect drug	Atorvastatin 40 mg (interacting drug: clarithromycin)	Yes
Adverse event	Rhabdomyolysis with acute kidney injury	Yes

Conclusion: valid ICSR. Clock starts on 12 Oct 2026 (Day 0).

2. Case Narrative

A 58-year-old male with type 2 diabetes, hypertension and dyslipidaemia, on stable atorvastatin 40 mg daily since August 2024 (plus metformin, amlodipine and lisinopril), was prescribed clarithromycin 500 mg twice daily on 22 Sep 2026 by a general practitioner for community-acquired pneumonia. On 27 Sep (clarithromycin day 6) he developed diffuse myalgia and proximal muscle weakness; on 28 Sep he noticed dark, tea-coloured urine with reduced output. He was admitted on 30 Sep 2026. Investigations showed CK 18,400 U/L (ULN 190), creatinine 2.9 mg/dL (baseline 1.0), potassium 5.8 mmol/L, AST 410 U/L, ALT 190 U/L and positive urine myoglobin. TSH was normal. There was no trauma, seizure, strenuous exercise or alcohol use. Rhabdomyolysis with acute kidney injury was diagnosed. Atorvastatin, clarithromycin, metformin and lisinopril were stopped; pneumonia was completed on an alternative antibiotic. He received IV hydration; dialysis was not required. By discharge on 6 Oct 2026 CK was 480 U/L, creatinine 1.3 mg/dL and symptoms had largely resolved. The physician considered the event related to the atorvastatin-clarithromycin combination. Outcome at report: recovering. No rechallenge.

3. Event Timeline

Date	Clarithromycin day	Event
Aug 2024	-	Atorvastatin 40 mg daily started; tolerated for about 2 years without muscle symptoms
22 Sep 2026	Day 1	Clarithromycin 500 mg BID started for pneumonia (different prescriber)
27 Sep 2026	Day 6	Onset of myalgia and proximal weakness
28 Sep 2026	Day 7	Dark urine, reduced urine output
30 Sep 2026	Day 9	Admitted. CK 18,400, creatinine 2.9, K 5.8. Suspect drugs stopped; IV fluids
2 Oct 2026	-	CK 6,200; creatinine 2.2; K 4.9
6 Oct 2026	-	Discharged. CK 480; creatinine 1.3; hospital stay 6 days
12 Oct 2026	-	Report received by MAH (regulatory Day 0)

4. Drug Information

Drug	Role	Dose / route	Dates	Indication	Action, dechallenge
Atorvastatin	Suspect	40 mg OD oral	Aug 2024 to 30 Sep 2026	Dyslipidaemia	Withdrawn; positive dechallenge; no rechallenge
Clarithromycin	Interacting	500 mg BID oral	22 Sep to 30 Sep 2026	Pneumonia	Withdrawn; positive dechallenge
Metformin	Concomitant	1000 mg BID	Ongoing (long-term)	T2DM	Held for AKI
Lisinopril	Concomitant	10 mg OD	Ongoing (long-term)	Hypertension	Held for AKI
Amlodipine	Concomitant	5 mg OD	Ongoing (long-term)	Hypertension	Continued

Interaction mechanism: clarithromycin is a strong CYP3A4 inhibitor (it also inhibits P-gp and OATP1B1), which substantially raises atorvastatin exposure. The atorvastatin label advises limiting the dose to 20 mg daily with clarithromycin. This patient was taking 40 mg. Amlodipine is a weak CYP3A4 inhibitor, but it had been co-administered stably for 2 years, so its contribution is considered minor.

5. ADR Identification and MedDRA Coding

Event (PT)	SOC	Seriousness	Expectedness (atorvastatin RSI)
Rhabdomyolysis	Musculoskeletal and connective tissue disorders	Serious (hospitalisation, medically important)	Expected (warnings, adverse reactions)
Acute kidney injury	Renal and urinary disorders	Serious (hospitalisation)	Expected (secondary to rhabdomyolysis)
Hyperkalaemia	Metabolism and nutrition disorders	Non-serious as standalone (K 5.8, no ECG change)	Not listed separately; recognised complication of rhabdomyolysis
Blood creatine phosphokinase increased	Investigations	Supportive laboratory finding	Expected
Myoglobinuria	Renal and urinary disorders	Supportive finding	Expected (feature of rhabdomyolysis)
Transaminases increased	Investigations	Non-serious	Expected (labeled)
Drug interaction	General disorders and administration site conditions	Circumstance, not an outcome	Expected (interaction labeled)

Myalgia, weakness and dark urine are not coded separately because a unifying diagnosis was reported. Use the current MedDRA version when coding.

6. Seriousness Assessment (ICH E2A)

Criterion	Met?	Basis
Death	No	Patient alive
Life-threatening	No	No ICU, no dialysis, no arrhythmia; K 5.8 only
Hospitalisation (initial or prolonged)	**Yes**	Admitted 30 Sep to 6 Oct 2026 (6 days)
Persistent or significant disability	No	Recovering; renal function near baseline
Congenital anomaly	N/A	-
Medically important event	**Yes**	Rhabdomyolysis with AKI is an important medical event

Overall: SERIOUS.

7. Severity Classification

Severity (intensity) is separate from seriousness. Here both are high.

Scale	Finding	Grade
CTCAE: CK increased	18,400 U/L, about 97 x ULN (>10 x ULN)	Grade 4
CTCAE: Acute kidney injury	Hospitalisation indicated	Grade 3
KDIGO AKI staging	Creatinine 2.9 x baseline (2.0 to 2.9 x)	Stage 2
Clinical intensity	Unable to walk unaided; required admission and IV therapy	**Severe**

Confirm grading against the CTCAE version your organisation uses.

8. Causality Assessment

8.1 WHO-UMC

Drug	Time relationship	Alternative causes	Dechallenge	Category
Atorvastatin (suspect)	Plausible: onset on day 6 of the interacting drug	Interaction explains the event; no disease or other cause found (TSH normal, no exertion, no trauma)	Positive	**Probable / Likely**
Clarithromycin (interacting)	Plausible	Atorvastatin alone could contribute	Positive	**Possible**

8.2 Naranjo scale: atorvastatin

#	Question	Answer	Score
1	Previous conclusive reports of this reaction?	Yes (labeled)	+1
2	Event after the drug was given?	Yes	+2
3	Improved when the drug was stopped?	Yes	+1
4	Reappeared on rechallenge?	Not done	0
5	Alternative causes that could cause the event?	No (interaction treated as the mechanism)	+2
6	Reaction with placebo?	Not done	0
7	Toxic drug concentration in blood?	Not measured	0
8	Dose-response relationship?	Not assessable	0
9	Similar reaction on previous exposure?	No	0
10	Confirmed by objective evidence?	Yes (CK, myoglobinuria)	+1
	Total	**Probable (5 to 8)**	**7**

Clarithromycin (interacting drug): Naranjo total 4 (+1 prior reports, +2 temporal, +1 dechallenge, -1 alternative cause, +1 objective evidence) = **Possible**. Reporter causality is **related**; under ICH E2D the company may not downgrade it. Company causality: **probable**.

9. Preventability

Schumock-Thornton criteria: **preventable**. A documented drug interaction was involved, and the atorvastatin dose (40 mg) exceeded the labeled limit during clarithromycin use. Recommend prescriber education on interacting combinations.

10. Reporting Decision

Parameter	Determination
Source and territory	Domestic spontaneous HCP report, India, marketed product
Seriousness / expectedness	Serious / expected (listed in RSI)
Causality	Reporter: related. Company: probable
India (CDSCO / PvPI)	Expedited report for a serious case: 15 calendar days from Day 0. **Due 27 Oct 2026.** Include in the next PSUR
EU (if authorised)	All serious spontaneous cases to EudraVigilance within 15 days, regardless of expectedness
US (if marketed)	Serious and *expected*: not a 15-day alert; submit in periodic reports
Format	E2B(R3) where the regulator requires it; confirm the current requirements per authority

11. Follow-up Queries to the Reporter

- Product batch numbers and manufacturer for atorvastatin and clarithromycin.
- Exact atorvastatin start date and any recent dose change.
- Who prescribed clarithromycin, and was the interaction flagged at dispensing?
- Peak CK and creatinine values, urine output, ECG, and any dialysis consideration.
- Family or personal history of myopathy; thyroid, vitamin D and alcohol history.
- Date of full recovery, and whether a statin will be reintroduced (rechallenge is not needed for causality).

12. Signal and Risk Management View

Known, labeled risk; **no new signal**. Add the case to aggregate review of statin-macrolide interaction cases and monitor for clusters. Check that the interaction warning is prominent in the local prescribing information. The benefit-risk balance is unchanged. Clinical actions such as switching to a non-interacting antibiotic or pausing the statin are for the prescriber.

13. Summary

Final assessment	
Event	Rhabdomyolysis with acute kidney injury
Seriousness	Serious (hospitalisation, medically important)
Expectedness	Expected (listed)
Severity	Severe (CTCAE CK Grade 4, AKI Grade 3; KDIGO stage 2)
Causality	Atorvastatin: probable (Naranjo 7). Clarithromycin: possible (interacting)
Preventability	Preventable (known interaction, dose above label limit)
Outcome	Recovering
Reporting	Expedited, 15 calendar days, due 27 Oct 2026
Signal	None new; routine aggregate monitoring